SAFETY DATA SHEET

according to Regulation (EC) No. 1907/2006

Creation Date 04-Oct-2010 **Revision Date** 11-Sep-2024 **Revision Number** 1

SECTION 1: IDENTIFICATION OF THE SUBSTANCE/MIXTURE AND OF THE COMPANY/UNDERTAKING

1.1. Product identifier

Product Description:	**Deblock, 3% TCA in Dichloromethane**
Cat No. :	**TS/0408/27SS**
Synonyms	Deblock Base 4

1.2. Relevant identified uses of the substance or mixture and uses advised against

Recommended Use	Laboratory chemicals.
Uses advised against	No Information available

1.3. Details of the supplier of the safety data sheet

Company .

EU entity/business name
Thermo Fisher Scientific
Janssen Pharmaceuticalaan 3a
2440 Geel, Belgium

UK entity/business name
Fisher Scientific UK
Bishop Meadow Road, Loughborough,
Leicestershire LE11 5RG, United Kingdom

Swiss distributor - Fisher Scientific AG
Neuhofstrasse 11, CH 4153 Reinach
Tel: +41 (0) 56 618 41 11
e-mail - infoch@thermofisher.com

E-mail address begel.sdsdesk@thermofisher.com

1.4. Emergency telephone number

Tel: 01509 231166
Chemtrec US: (800) 424-9300
Chemtrec EU: 001-703-527-3887

For customers in Switzerland:
Tox Info Suisse Emergency Number: **145 (24hr)**
Tox Info Suisse: +41-44 251 51 51 (Emergency number from abroad)
Chemtrec (24h) Toll-Free: 0800 564 402
Chemtrec Local: +41-43 508 20 11 (Zurich)

SECTION 2: HAZARDS IDENTIFICATION

2.1. Classification of the substance or mixture

CLP Classification - Regulation (EC) No 1272/2008

Physical hazards

Based on available data, the classification criteria are not met

Health hazards

Skin Corrosion/Irritation	Category 2 (H315)
Serious Eye Damage/Eye Irritation	Category 1 (H318)
Carcinogenicity	Category 2 (H351)
Specific target organ toxicity - (single exposure)	Category 3 (H335) (H336)

Environmental hazards

Chronic aquatic toxicity	Category 2 (H411)

Full text of Hazard Statements: see section 16

2.2. Label elements

Contains Dichloromethane & Trichloroacetic acid

Signal Word **Danger**

Hazard Statements

H315 - Causes skin irritation
H318 - Causes serious eye damage
H335 - May cause respiratory irritation
H336 - May cause drowsiness or dizziness
H351 - Suspected of causing cancer
H411 - Toxic to aquatic life with long lasting effects

Precautionary Statements

P280 - Wear protective gloves/protective clothing/eye protection/face protection
P302 + P352 - IF ON SKIN: Wash with plenty of soap and water
P332 + P313 - If skin irritation occurs: Get medical advice/attention
P304 + P340 - IF INHALED: Remove person to fresh air and keep comfortable for breathing
P305 + P351 + P338 - IF IN EYES: Rinse cautiously with water for several minutes. Remove contact lenses, if present and easy to do. Continue rinsing
P310 - Immediately call a POISON CENTER or doctor/physician

2.3. Other hazards

Substance is not considered persistent, bioaccumulative and toxic (PBT) / very persistent and very bioaccumulative (vPvB)

Toxic to terrestrial vertebrates
Contains a substance on the National Authorities Endocrine Disruptor Lists
Contains a known or suspected endocrine disruptor

SECTION 3: COMPOSITION/INFORMATION ON INGREDIENTS

3.2. Mixtures

Component	CAS No	EC No	Weight %	CLP Classification - Regulation (EC) No 1272/2008
Dichloromethane	75-09-2	EEC No. 200-838-9	97	Skin Irrit. 2 (H315) Eye Irrit. 2 (H319) STOT SE 3 (H336) Carc. 2 (H351)
Trichloroacetic acid	76-03-9	EEC No. 200-927-2	3	Skin Corr. 1A (H314) Eye Dam. 1 (H318) STOT SE 3 (H335) Aquatic Acute 1 (H400) Aquatic Chronic 1 (H410)

Component	Specific concentration limits (SCL's)	M-Factor	Component notes
Trichloroacetic acid	STOT SE 3 (H335) :: C>=1%	1	-

Components	Reach Registration Number	
Dichloromethane	01-2119480404-41	
Trichloroacetic acid	01-2119485186-30	

Full text of Hazard Statements: see section 16

SECTION 4: FIRST AID MEASURES

4.1. Description of first aid measures

General Advice If symptoms persist, call a physician.

Eye Contact Rinse immediately with plenty of water, also under the eyelids, for at least 15 minutes. Get medical attention.

Skin Contact Wash off immediately with plenty of water for at least 15 minutes. If skin irritation persists, call a physician.

Ingestion Clean mouth with water and drink afterwards plenty of water.

Inhalation Remove to fresh air. If not breathing, give artificial respiration. Get medical attention if symptoms occur.

Self-Protection of the First Aider Ensure that medical personnel are aware of the material(s) involved, take precautions to protect themselves and prevent spread of contamination.

4.2. Most important symptoms and effects, both acute and delayed

Difficulty in breathing. Causes severe eye damage. Causes eye burns. Inhalation of high vapor concentrations may cause symptoms like headache, dizziness, tiredness, nausea and vomiting

4.3. Indication of any immediate medical attention and special treatment needed

Notes to Physician A patient adversely affected by exposure to this product should not be given adrenaline (epinephrine) or similar heart stimulant since these would increase the risk of cardiac arrhythmias. Treat symptomatically. Symptoms may be delayed.

SECTION 5: FIREFIGHTING MEASURES

5.1. Extinguishing media

Suitable Extinguishing Media
Water spray, carbon dioxide (CO2), dry chemical, alcohol-resistant foam.

Extinguishing media which must not be used for safety reasons
No information available.

5.2. Special hazards arising from the substance or mixture

Thermal decomposition can lead to release of irritating gases and vapors.

Hazardous Combustion Products
Phosgene, Chlorine, Hydrocarbons, Hydrogen chloride gas.

5.3. Advice for firefighters

As in any fire, wear self-contained breathing apparatus pressure-demand, MSHA/NIOSH (approved or equivalent) and full protective gear.

SECTION 6: ACCIDENTAL RELEASE MEASURES

6.1. Personal precautions, protective equipment and emergency procedures

Use personal protective equipment as required. Ensure adequate ventilation.

6.2. Environmental precautions

Do not flush into surface water or sanitary sewer system. See Section 12 for additional Ecological Information. Avoid release to the environment. Collect spillage.

6.3. Methods and material for containment and cleaning up

Soak up with inert absorbent material. Keep in suitable, closed containers for disposal.

6.4. Reference to other sections

Refer to protective measures listed in Sections 8 and 13.

SECTION 7: HANDLING AND STORAGE

7.1. Precautions for safe handling

Wear personal protective equipment/face protection. Do not get in eyes, on skin, or on clothing. Ensure adequate ventilation. Avoid ingestion and inhalation.

Hygiene Measures
Handle in accordance with good industrial hygiene and safety practice. Keep away from food, drink and animal feeding stuffs. Do not eat, drink or smoke when using this product. Remove and wash contaminated clothing and gloves, including the inside, before re-use. Wash hands before breaks and after work.

7.2. Conditions for safe storage, including any incompatibilities

Keep containers tightly closed in a dry, cool and well-ventilated place. Corrosives area.

Technical Rules for Hazardous Substances (TRGS) 510 Storage Class (LGK) (Germany)	Storage Class/LGK 10
Switzerland - Storage of hazardous substances	Storage class - SC 10/12 https://www.kvu.ch/de/themen/stoffe-und-produkte https://www.kvu.ch/fr/themes/substances-et-produits https://www.kvu.ch/it/temi/sostanze-e-prodotti

7.3. Specific end use(s)

Use in laboratories

SECTION 8: EXPOSURE CONTROLS/PERSONAL PROTECTION

8.1. Control parameters

Exposure limits

List source(s): **UK** - EH40/2005 Work Exposure Limits, Forth edition. Published 2020. **IRE -** 2010 Code of Practice for the Safety, Health and Welfare at Work (Chemical Agents) Regulations 2001. Published by the Health and Safety Authority. **CH** - The Government of Switzerland has set a directive on limit values for working materials (Grenzwerte am Arbeitsplatz) which is based on the Swiss Federal Regulation "Verordnung über die Verhütung von Unfällen und Berufskrankheiten". This directive is administered, periodically revised and enforced by SUVA (Swiss National Accident Insurance Fund). **EU** - Commission Directive (EU) 2019/1831 of 24 October 2019 establishing a fifth list of indicative occupational exposure limit values pursuant to Council Directive 98/24/EC and amending Commission Directive 2000/39/EC

Component	European Union	The United Kingdom	France	Belgium	Spain
Dichloromethane	TWA: 353 mg/m³ (8h) TWA: 100 ppm (8h) STEL: 706 mg/m³ (15min) STEL: 200 ppm (15min) Skin	STEL: 200 ppm 15 min STEL: 706 mg/m³ 15 min TWA: 353 mg/m³ 8 hr TWA: 100 ppm 8 hr Skin	TWA / VME: 50 ppm (8 heures). restrictive limit TWA / VME: 178 mg/m³ (8 heures). restrictive limit STEL / VLCT: 100 ppm. restrictive limit STEL / VLCT: 356 mg/m³. restrictive limit Peau	TWA: 50 ppm 8 uren TWA: 177 mg/m³ 8 uren STEL: 200 ppm 15 minuten STEL: 706 mg/m³ 15 minuten Huid	STEL / VLA-EC: 100 ppm (15 minutos). STEL / VLA-EC: 353 mg/m³ (15 minutos). TWA / VLA-ED: 50 ppm (8 horas) TWA / VLA-ED: 177 mg/m³ (8 horas)
Trichloroacetic acid			TWA / VME: 1 ppm (8 heures). TWA / VME: 5 mg/m³ (8 heures).	TWA: 1 ppm 8 uren TWA: 6.8 mg/m³ 8 uren	TWA / VLA-ED: 1 ppm (8 horas) TWA / VLA-ED: 6.8 mg/m³ (8 horas)

Component	Italy	Germany	Portugal	The Netherlands	Finland
Dichloromethane	TWA: 175 mg/m³ 8 ore. Time Weighted Average TWA: 50 ppm 8 ore. Time Weighted Average STEL: 353 mg/m³ 15 minuti. Short-term STEL: 100 ppm 15 minuti. Short-term Pelle	TWA: 50 ppm (8 Stunden). AGW - exposure factor 2 TWA: 180 mg/m³ (8 Stunden). AGW - exposure factor 2 TWA: 50 ppm (8 Stunden). MAK TWA: 180 mg/m³ (8 Stunden). MAK Höhepunkt: 100 ppm Höhepunkt: 360 mg/m³ Haut	STEL: 706 mg/m³ 15 minutos STEL: 200 ppm 15 minutos TWA: 353 mg/m³ 8 horas TWA: 100 ppm 8 horas Pele	huid STEL: 200 ppm 15 minuten STEL: 706 mg/m³ 15 minuten TWA: 100 ppm 8 uren TWA: 353 mg/m³ 8 uren	TWA: 50 ppm 8 tunteina TWA: 177 mg/m³ 8 tunteina STEL: 100 ppm 15 minuutteina STEL: 353 mg/m³ 15 minuutteina Iho
Trichloroacetic acid		TWA: 0.2 ppm (8 Stunden). AGW - exposure factor 1 TWA: 1.4 mg/m³ (8 Stunden). AGW -	TWA: 0.5 ppm 8 horas		

		exposure factor 1 TWA: 0.2 ppm (8 Stunden). MAK can occur as vapor and aerosol at the same time TWA: 1.4 mg/m³ (8 Stunden). MAK can occur as vapor and aerosol at the same time Höhepunkt: 0.2 ppm Höhepunkt: 1.4 mg/m³			

Component	Austria	Denmark	Switzerland	Poland	Norway
Dichloromethane	Haut MAK-KZGW: 200 ppm 15 Minuten MAK-KZGW: 700 mg/m³ 15 Minuten MAK-TMW: 50 ppm 8 Stunden MAK-TMW: 175 mg/m³ 8 Stunden	TWA: 35 ppm 8 timer TWA: 122 mg/m³ 8 timer STEL: 706 mg/m³ 15 minutter STEL: 200 ppm 15 minutter Hud	Haut/Peau STEL: 200 ppm 15 Minuten STEL: 706 mg/m³ 15 Minuten TWA: 50 ppm 8 Stunden TWA: 177 mg/m³ 8 Stunden	STEL: 353 mg/m³ 15 minutach TWA: 88 mg/m³ 8 godzinach	TWA: 15 ppm 8 timer TWA: 50 mg/m³ 8 timer STEL: 45 ppm 15 minutter. value from the regulation STEL: 150 mg/m³ 15 minutter. value from the regulation Hud
Trichloroacetic acid	MAK-TMW: 1 ppm 8 Stunden MAK-TMW: 5 mg/m³ 8 Stunden	TWA: 1 mg/m³ 8 timer STEL: 2 mg/m³ 15 minutter	TWA: 1 ppm 8 Stunden TWA: 7 mg/m³ 8 Stunden	STEL: 4 mg/m³ 15 minutach TWA: 2 mg/m³ 8 godzinach	TWA: 0.75 ppm 8 timer TWA: 5 mg/m³ 8 timer STEL: 2.25 ppm 15 minutter. value calculated STEL: 10 mg/m³ 15 minutter. value calculated

Component	Bulgaria	Croatia	Ireland	Cyprus	Czech Republic
Dichloromethane	TWA: 353 mg/m³ TWA: 100 ppm STEL : 706 mg/m³ STEL : 200 ppm Skin notation	kože TWA-GVI: 100 ppm 8 satima. TWA-GVI: 353 mg/m³ 8 satima. STEL-KGVI: 200 ppm 15 minutama. STEL-KGVI: 706 mg/m³ 15 minutama.	TWA: 100 ppm 8 hr. TWA: 353 mg/m³ 8 hr. STEL: 200 ppm 15 min STEL: 706 mg/m³ 15 min Skin	Skin-potential for cutaneous absorption STEL: 706 mg/m³ STEL: 200 ppm TWA: 353 mg/m³ TWA: 100 ppm	TWA: 200 mg/m³ 8 hodinách. Potential for cutaneous absorption Ceiling: 500 mg/m³
Trichloroacetic acid	TWA: 7.0 mg/m³		TWA: 0.5 ppm 8 hr. STEL: 1.5 ppm 15 min		

Component	Estonia	Gibraltar	Greece	Hungary	Iceland
Dichloromethane	Nahk TWA: 35 ppm 8 tundides. TWA: 120 mg/m³ 8 tundides. STEL: 70 ppm 15 minutites. STEL: 250 mg/m³ 15 minutites.	Skin notation TWA: 353 mg/m³ 8 hr TWA: 100 ppm 8 hr STEL: 706 mg/m³ 15 min STEL: 200 ppm 15 min	skin - potential for cutaneous absorption STEL: 200 ppm STEL: 706 mg/m³ TWA: 100 ppm TWA: 353 mg/m³	STEL: 200 ppm 15 percekben. CK STEL: 706 mg/m³ 15 percekben. CK TWA: 100 ppm 8 órában. AK TWA: 353 mg/m³ 8 órában. AK lehetséges borön keresztüli felszívódás	TWA: 35 ppm 8 klukkustundum. TWA: 122 mg/m³ 8 klukkustundum. Skin notation Ceiling: 70 ppm Ceiling: 244 mg/m³
Trichloroacetic acid					TWA: 1 mg/m³ 8 klukkustundum. Ceiling: 2 mg/m³

Component	Latvia	Lithuania	Luxembourg	Malta	Romania
Dichloromethane	skin - potential for cutaneous exposure STEL: 150 mg/m³ STEL: 42 ppm TWA: 120 mg/m³ TWA: 34 ppm	TWA: 35 ppm IPRD TWA: 120 mg/m³ IPRD Oda STEL: 70 ppm STEL: 250 mg/m³	Possibility of significant uptake through the skin TWA: 100 ppm 8 Stunden TWA: 353 mg/m³ 8 Stunden	possibility of significant uptake through the skin TWA: 100 ppm TWA: 353 mg/m³ STEL: 200 ppm 15 minuti	Skin notation TWA: 100 ppm 8 ore TWA: 353 mg/m³ 8 ore STEL: 200 ppm 15 minute STEL: 706 mg/m³ 15

			STEL: 200 ppm 15 Minuten STEL: 706 mg/m³ 15 Minuten	STEL: 706 mg/m³ 15 minuti	minute
Trichloroacetic acid	TWA: 5 mg/m³				

Component	Russia	Slovak Republic	Slovenia	Sweden	Turkey
Dichloromethane	TWA: 50 mg/m³ 0922 MAC: 100 mg/m³	Ceiling: 706 mg/m³ Potential for cutaneous absorption TWA: 100 ppm TWA: 353 mg/m³	TWA: 100 ppm 8 urah TWA: 353 mg/m³ 8 urah Koža STEL: 200 ppm 15 minutah STEL: 706 mg/m³ 15 minutah	Binding STEL: 70 ppm 15 minuter Binding STEL: 250 mg/m³ 15 minuter TLV: 35 ppm 8 timmar. NGV TLV: 120 mg/m³ 8 timmar. NGV Hud	
Trichloroacetic acid	Skin notation MAC: 5 mg/m³		TWA: 1.4 mg/m³ 8 urah TWA: 0.2 ppm 8 urah STEL: 0.2 ppm 15 minutah STEL: 1.4 mg/m³ 15 minutah		

Biological limit values

List source(s): **UK** - Biological Monitoring Guidance Values provided by the UK's Health and Safety Executive (HSE) Control of Substances Hazardous to Health Regulations (COSHH) 2002 (as amended) and EH40/2005.

Component	European Union	United Kingdom	France	Spain	Germany
Dichloromethane		Carbon monoxide: 30 ppm end-tidal breath post shift	Dichloromethane: 0.2 mg/L urine end of shift Carboxyhémoglobine sanguine: 3.5 % blood end of shift	Dichloromethane: 0.3 mg/L urine end of shift	Dichloromethane: 500 µg/L whole blood (immediately after exposure)

Component	Italy	Finland	Denmark	Bulgaria	Romania
Dichloromethane					Carboxyhemoglobin: 5 % Hemoglobin blood end of shift Methylene chloride: 0.3 mg/L urine end of shift Methylene chloride: 1 mg/L blood end of shift

Component	Gibraltar	Latvia	Slovak Republic	Luxembourg	Turkey
Dichloromethane			Dichloromethane: 1 mg/L blood end of exposure or work shift Carboxyhemoglobin: 5 % of hemoglobin blood end of exposure or work shift		

Monitoring methods

BS EN 14042:2003 Title Identifier: Workplace atmospheres. Guide for the application and use of procedures for the assessment of exposure to chemical and biological agents.
MDHS70 General methods for sampling airborne gases and vapours
MDHS 88 Volatile organic compounds in air. Laboratory method using diffusive samplers, solvent desorption and gas chromatography
MDHS 96 Volatile organic compounds in air - Laboratory method using pumped solid sorbent tubes, solvent desorption and gas chromatography

Derived No Effect Level (DNEL) / Derived Minimum Effect Level (DMEL)

See values below:

Component	Acute effects local	Acute effects	Chronic effects local	Chronic effects

	(Oral)	systemic (Oral)	(Oral)	systemic (Oral)
Trichloroacetic acid 76-03-9 (3)				0.7 mg/kg/d

Component	Acute effects local (Dermal)	Acute effects systemic (Dermal)	Chronic effects local (Dermal)	Chronic effects systemic (Dermal)
Dichloromethane 75-09-2 (97)				DNEL = 12mg/kg bw/day
Trichloroacetic acid 76-03-9 (3)	DMEL = 5% in mixture (weight basis)	DNEL = 1.41mg/kg bw/day		DNEL = 1.41mg/kg bw/day

Component	Acute effects local (Inhalation)	Acute effects systemic (Inhalation)	Chronic effects local (Inhalation)	Chronic effects systemic (Inhalation)
Dichloromethane 75-09-2 (97)		DMEL = 132.14mg/m³		DNEL = 176mg/m³
Trichloroacetic acid 76-03-9 (3)		DNEL = 124.3mg/m³		DNEL = 124.3mg/m³

Predicted No Effect Concentration (PNEC)
See values below.

Component	Fresh water	Fresh water sediment	Water Intermittent	Microorganisms in sewage treatment	Soil (Agriculture)
Dichloromethane 75-09-2 (97)	PNEC = 130µg/L PNEC = 0.31mg/L	PNEC = 163µg/kg sediment dw PNEC = 2.57mg/kg sediment dw	PNEC = 0.27mg/L	PNEC = 26mg/L	PNEC = 173µg/kg soil dw PNEC = 0.33mg/kg soil dw
Trichloroacetic acid 76-03-9 (3)	PNEC = 0.17µg/L	PNEC = 0.143µg/kg sediment dw	PNEC = 2.7µg/L	PNEC = 100mg/L	PNEC = 4.6µg/kg soil dw

Component	Marine water	Marine water sediment	Marine water Intermittent	Food chain	Air
Dichloromethane 75-09-2 (97)	PNEC = 130µg/L PNEC = 0.031mg/L	PNEC = 163µg/kg sediment dw PNEC = 0.26mg/kg sediment dw	PNEC = 0.027mg/L		
Trichloroacetic acid 76-03-9 (3)	PNEC = 0.017µg/L	PNEC = 0.0143µg/kg sediment dw		PNEC = 23.5mg/kg food	

8.2. Exposure controls

Engineering Measures
Use only under a chemical fume hood. Ensure that eyewash stations and safety showers are close to the workstation location.
Ensure adequate ventilation, especially in confined areas.
Wherever possible, engineering control measures such as the isolation or enclosure of the process, the introduction of process or equipment changes to minimise release or contact, and the use of properly designed ventilation systems, should be adopted to control hazardous materials at source

Personal protective equipment

Eye Protection Goggles (European standard - EN 166)

Hand Protection Protective gloves

Glove material	Breakthrough time	Glove thickness	EU standard	Glove comments
Viton (R)	< 120 minutes	0.7 mm	EN 374	As tested under EN374-3 Determination of
Nitrile rubber	< 4 minutes	0.38 mm		Resistance to Permeation by Chemicals
PVA	> 360 minutes			

Skin and body protection Long sleeved clothing.

Inspect gloves before use. observe the instructions regarding permeability and breakthrough time which are provided by the supplier of the gloves. (Refer to manufacturer/supplier for information) gloves are suitable for the task: Chemical compatability, Dexterity, Operational conditions, User susceptibility, e.g. sensitisation effects, also take into consideration the specific local conditions under which the product is used, such as the danger of cuts, abrasion. gloves with care avoiding skin contamination.

Respiratory Protection When workers are facing concentrations above the exposure limit they must use appropriate certified respirators.
To protect the wearer, respiratory protective equipment must be the correct fit and be used and maintained properly

Large scale/emergency use Use a NIOSH/MSHA or European Standard EN 136 approved respirator if exposure limits are exceeded or if irritation or other symptoms are experienced
Recommended Filter type: low boiling organic solvent Type AX Brown conforming to EN371

Small scale/Laboratory use Use a NIOSH/MSHA or European Standard EN 149:2001 approved respirator if exposure limits are exceeded or if irritation or other symptoms are experienced.
Recommended half mask:- Valve filtering: EN405; or; Half mask: EN140; plus filter, EN 141
When RPE is used a face piece Fit Test should be conducted

Environmental exposure controls Prevent product from entering drains. Do not allow material to contaminate ground water system.

SECTION 9: PHYSICAL AND CHEMICAL PROPERTIES

9.1. Information on basic physical and chemical properties

Physical State	Liquid	
Appearance	Colorless	
Odor	sweet	
Odor Threshold	No data available	
Melting Point/Range	-97 °C / -142.6 °F	
Softening Point	No data available	
Boiling Point/Range	40 °C / 104 °F	
Flammability (liquid)	No data available	
Flammability (solid,gas)	Not applicable	Liquid
Explosion Limits	**Lower** 13 vol% **Upper** 22 vol%	
Flash Point	Not applicable	**Method -** No information available
Autoignition Temperature	556 °C / 1032.8 °F	
Decomposition Temperature	> 120°C	
pH	Not applicable	
Viscosity	0.43 cP at 20 °C	
Water Solubility	Immiscible	
Solubility in other solvents	No information available	
Partition Coefficient (n-octanol/water)		
Component	**log Pow**	
Dichloromethane	1.25	
Trichloroacetic acid	1,44	
Vapor Pressure	350 mbar @ 20 °C	
Density / Specific Gravity	1.325	
Bulk Density	Not applicable	Liquid
Vapor Density	No data available	(Air = 1.0)
Particle characteristics	Not applicable (liquid)	

9.2. Other information

SECTION 10: STABILITY AND REACTIVITY

10.1. Reactivity

None known, based on information available

10.2. Chemical stability

Stable under normal conditions.

10.3. Possibility of hazardous reactions

Hazardous Polymerization Hazardous polymerization does not occur.
Hazardous Reactions None under normal processing.

10.4. Conditions to avoid

Incompatible products. Excess heat.

10.5. Incompatible materials

Strong oxidizing agents. Strong acids. Bases. Alcohols. Amines. Metals.

10.6. Hazardous decomposition products

Phosgene. Chlorine. Hydrocarbons. Hydrogen chloride gas.

SECTION 11: TOXICOLOGICAL INFORMATION

11.1. Information on hazard classes as defined in Regulation (EC) No 1272/2008

Product Information No acute toxicity information is available for this product

(a) acute toxicity;
Oral Based on ATE data, the classification criteria are not met
Dermal Based on ATE data, the classification criteria are not met
Inhalation Based on ATE data, the classification criteria are not met

Toxicology data for the components

Component	LD50 Oral	LD50 Dermal	LC50 Inhalation
Dichloromethane	> 2000 mg/kg (Rat)	> 2000 mg/kg (Rat)	53 mg/L (Rat) 6 h 76000 mg/m³ (Rat) 4 h
Trichloroacetic acid	3320 mg/kg rat	LD50 > 2000 mg/kg (Rat)	-

(b) skin corrosion/irritation; Category 2

(c) serious eye damage/irritation; Category 1

(d) respiratory or skin sensitization;
Respiratory Based on available data, the classification criteria are not met
Skin Based on available data, the classification criteria are not met

(e) germ cell mutagenicity; Based on available data, the classification criteria are not met

(f) carcinogenicity; Category 2

The table below indicates whether each agency has listed any ingredient as a carcinogen

Component	EU	UK	Germany	IARC
Dichloromethane				Group 2A
Trichloroacetic acid				Group 2B

(g) reproductive toxicity; Based on available data, the classification criteria are not met

(h) STOT-single exposure; Category 3

Results / Target organs Central nervous system (CNS), Respiratory system.

(i) STOT-repeated exposure; Based on available data, the classification criteria are not met

Target Organs No information available.

(j) aspiration hazard; Based on available data, the classification criteria are not met

Symptoms / effects,both acute and delayed Inhalation of high vapor concentrations may cause symptoms like headache, dizziness, tiredness, nausea and vomiting.

11.2. Information on other hazards

Endocrine Disrupting Properties .
Assess endocrine disrupting properties for human health Contains a substance on the National Authorities Endocrine Disruptor Lists

SECTION 12: ECOLOGICAL INFORMATION

12.1. Toxicity

Ecotoxicity effects Toxic to aquatic organisms, may cause long-term adverse effects in the aquatic environment. The product contains following substances which are hazardous for the environment.

Component	Freshwater Fish	Water Flea	Freshwater Algae
Dichloromethane	Pimephales promelas: LC50:193 mg/L/96h	EC50: 140 mg/L/48h	EC50:>660 mg/L/96h
Trichloroacetic acid	>277 mg/l	110 mg/l	0.27 mg/l

Component	Microtox	M-Factor
Dichloromethane	EC50: 1 mg/L/24 h EC50: 2.88 mg/L/15 min	
Trichloroacetic acid		1

12.2. Persistence and degradability

Persistence Persistence is unlikely, based on information available, Immiscible with water.
Degradation in sewage treatment plant Contains substances known to be hazardous to the environment or not degradable in waste water treatment plants.

12.3. Bioaccumulative potential

May have some potential to bioaccumulate

Component	log Pow	Bioconcentration factor (BCF)
Dichloromethane	1.25	6.4 - 40 dimensionless
Trichloroacetic acid	1,44	0.4-1.7 Cyprinus caprio

12.4. Mobility in soil	The product contains volatile organic compounds (VOC) which will evaporate easily from all surfaces. Spillage unlikely to penetrate soil. The product is insoluble and sinks in water. Will likely be mobile in the environment due to its volatility. Disperses rapidly in air
12.5. Results of PBT and vPvB assessment	Substance is not considered persistent, bioaccumulative and toxic (PBT) / very persistent and very bioaccumulative (vPvB).
12.6. Endocrine disrupting properties	
Endocrine Disruptor Information	This product does not contain any known or suspected endocrine disruptors
12.7. Other adverse effects	
Persistent Organic Pollutant	This product does not contain any known or suspected substance
Ozone Depletion Potential	This product does not contain any known or suspected substance

SECTION 13: DISPOSAL CONSIDERATIONS

13.1. Waste treatment methods

Waste from Residues/Unused Products	Waste is classified as hazardous. Dispose of in accordance with federal, state and local regulations. Dispose of in accordance with the European Directives on waste and hazardous waste. Dispose of in accordance with local regulations.
Contaminated Packaging	Do not reuse empty containers. Dispose of in accordance with local regulations. Dispose of this container to hazardous or special waste collection point.
European Waste Catalogue (EWC)	According to the European Waste Catalog, Waste Codes are not product specific, but application specific.
Other Information	Do not flush to sewer. Waste codes should be assigned by the user based on the application for which the product was used. Do not empty into drains. Do not let this chemical enter the environment.
Switzerland - Waste Ordinance	Disposal should be in accordance with applicable regional, national and local laws and regulations. Ordinance on the Avoidance and the Disposal of Waste (Waste Ordinance, ADWO) SR 814.600 https://www.fedlex.admin.ch/eli/cc/2015/891/en

SECTION 14: TRANSPORT INFORMATION

IMDG/IMO

14.1. UN number	UN2922
14.2. UN proper shipping name	Corrosive liquid, toxic, n.o.s.
Technical Shipping Name	Trichloroacetic acid, Dichloromethane
14.3. Transport hazard class(es)	8
Subsidiary Hazard Class	6.1
14.4. Packing group	III

ADR

14.1. UN number	UN2922
14.2. UN proper shipping name	Corrosive liquid, toxic, n.o.s.
Technical Shipping Name	Trichloroacetic acid, Dichloromethane
14.3. Transport hazard class(es)	8

Subsidiary Hazard Class	6.1
14.4. Packing group	III

IATA

14.1. UN number	UN2922
14.2. UN proper shipping name	Corrosive liquid, toxic, n.o.s.
Technical Shipping Name	Trichloroacetic acid, Dichloromethane
14.3. Transport hazard class(es)	8
Subsidiary Hazard Class	6.1
14.4. Packing group	III

14.5. Environmental hazards	Dangerous for the environment Product is a marine pollutant according to the criteria set by IMDG/IMO
14.6. Special precautions for user	No special precautions required.
14.7. Maritime transport in bulk according to IMO instruments	Not applicable, packaged goods

SECTION 15: REGULATORY INFORMATION

15.1. Safety, health and environmental regulations/legislation specific for the substance or mixture

International Inventories

Europe (EINECS/ELINCS/NLP), China (IECSC), Taiwan (TCSI), Korea (KECL), Japan (ENCS), Japan (ISHL), Canada (DSL/NDSL), Australia (AICS), New Zealand (NZIoC), Philippines (PICCS). US EPA (TSCA) - Toxic Substances Control Act, (40 CFR Part 710)

Component	CAS No	EINECS	ELINCS	NLP	IECSC	TCSI	KECL	ENCS	ISHL
Dichloromethane	75-09-2	200-838-9	-	-	X	X	KE-23893	X	X
Trichloroacetic acid	76-03-9	200-927-2	-	-	X	X	KE-34058	X	X

Component	CAS No	TSCA	TSCA Inventory notification - Active-Inactive	DSL	NDSL	AICS	NZIoC	PICCS
Dichloromethane	75-09-2	X	ACTIVE	X	-	X	X	X
Trichloroacetic acid	76-03-9	X	ACTIVE	X	-	X	X	X

Legend: X - Listed '-' - Not Listed **KECL** - NIER number or KE number (http://ncis.nier.go.kr/en/main.do)

Authorisation/Restrictions according to EU REACH

Component	CAS No	REACH (1907/2006) - Annex XIV - Substances Subject to Authorization	REACH (1907/2006) - Annex XVII - Restrictions on Certain Dangerous Substances	REACH Regulation (EC 1907/2006) article 59 - Candidate List of Substances of Very High Concern (SVHC)
Dichloromethane	75-09-2	-	Use restricted. See entry 59. (see link for restriction details) Use restricted. See entry 75. (see link for restriction details)	-
Trichloroacetic acid	76-03-9	-	Use restricted. See entry 75. (see link for restriction details)	-

REACH links
https://echa.europa.eu/substances-restricted-under-reach

Seveso III Directive (2012/18/EC)

Component	CAS No	Seveso III Directive (2012/18/EC) - Qualifying Quantities for Major Accident Notification	Seveso III Directive (2012/18/EC) - Qualifying Quantities for Safety Report Requirements
Dichloromethane	75-09-2	Not applicable	Not applicable
Trichloroacetic acid	76-03-9	Not applicable	Not applicable

Regulation (EC) No 649/2012 of the European Parliament and of the Council of 4 July 2012 concerning the export and import of dangerous chemicals
Not applicable

Contains component(s) that meet a 'definition' of per & poly fluoroalkyl substance (PFAS)?
Not applicable

Take note of Directive 98/24/EC on the protection of the health and safety of workers from the risks related to chemical agents at work .
Take note of Directive 2000/39/EC establishing a first list of indicative occupational exposure limit values

National Regulations

UK - Take note of Control of Substances Hazardous to Health Regulations (COSHH) 2002 and 2005 Amendment

WGK Classification Water endangering class = 2 (self classification)

Component	Germany - Water Classification (AwSV)	Germany - TA-Luft Class
Dichloromethane	WGK2	Class I : 20 mg/m³ (Massenkonzentration)
Trichloroacetic acid	WGK2	Class I : 20 mg/m³ (Massenkonzentration)

Component	France - INRS (Tables of occupational diseases)
Dichloromethane	Tableaux des maladies professionnelles (TMP) - RG 12

Swiss Regulations
Article 4 para. 4 of the Ordinance on the protection of young people in the workplace (SR 822.115) and Article 1 lit. f of the EAER regulation on hazardous work and young people (SR 822.115.2).
Take note on Article 13 Maternity Ordinance (SR 822.111.52) with regards expectant and nursing mothers.

Component	Switzerland - Ordinance on the Reduction of Risk from handling of hazardous substances preparation (SR 814.81)	Switzerland - Ordinance on Incentive Taxes on Volatile Organic Compounds (OVOC)	Switzerland - Ordinance of the Rotterdam Convention on the Prior Informed Consent Procedure
Dichloromethane 75-09-2 (97)	Persistent Organic Pollutants (POPs) Prohibited and Restricted Substances	Group I	

15.2. Chemical safety assessment

Chemical Safety Assessment/Reports (CSA/CSR) are not required for mixtures

SECTION 16: OTHER INFORMATION

Full text of H-Statements referred to under sections 2 and 3

H314 - Causes severe skin burns and eye damage
H315 - Causes skin irritation
H318 - Causes serious eye damage
H319 - Causes serious eye irritation
H335 - May cause respiratory irritation
H336 - May cause drowsiness or dizziness
H351 - Suspected of causing cancer
H400 - Very toxic to aquatic life
H410 - Very toxic to aquatic life with long lasting effects

Legend

CAS - Chemical Abstracts Service
EINECS/ELINCS - European Inventory of Existing Commercial Chemical Substances/EU List of Notified Chemical Substances
PICCS - Philippines Inventory of Chemicals and Chemical Substances
IECSC - Chinese Inventory of Existing Chemical Substances
KECL - Korean Existing and Evaluated Chemical Substances
TSCA - United States Toxic Substances Control Act Section 8(b) Inventory
DSL/NDSL - Canadian Domestic Substances List/Non-Domestic Substances List
ENCS - Japanese Existing and New Chemical Substances
AICS - Australian Inventory of Chemical Substances
NZIoC - New Zealand Inventory of Chemicals

WEL - Workplace Exposure Limit
ACGIH - American Conference of Governmental Industrial Hygienists
DNEL - Derived No Effect Level
RPE - Respiratory Protective Equipment
LC50 - Lethal Concentration 50%
NOEC - No Observed Effect Concentration
PBT - Persistent, Bioaccumulative, Toxic
TWA - Time Weighted Average
IARC - International Agency for Research on Cancer
Predicted No Effect Concentration (PNEC)
LD50 - Lethal Dose 50%
EC50 - Effective Concentration 50%
POW - Partition coefficient Octanol:Water
vPvB - very Persistent, very Bioaccumulative

ADR - European Agreement Concerning the International Carriage of Dangerous Goods by Road
IMO/IMDG - International Maritime Organization/International Maritime Dangerous Goods Code
OECD - Organisation for Economic Co-operation and Development
BCF - Bioconcentration factor
ICAO/IATA - International Civil Aviation Organization/International Air Transport Association
MARPOL - International Convention for the Prevention of Pollution from Ships
ATE - Acute Toxicity Estimate
VOC - (volatile organic compound)

Key literature references and sources for data
https://echa.europa.eu/information-on-chemicals
Suppliers safety data sheet, Chemadvisor - LOLI, Merck index, RTECS

Classification and procedure used to derive the classification for mixtures according to Regulation (EC) 1272/2008 [CLP]:

Physical hazards	On basis of test data
Health Hazards	Calculation method
Environmental hazards	Calculation method

Training Advice
Chemical hazard awareness training, incorporating labelling, Safety Data Sheets (SDS), Personal Protective Equipment (PPE) and hygiene.
First aid for chemical exposure, including the use of eye wash and safety showers.
Use of personal protective equipment, covering appropriate selection, compatibility, breakthrough thresholds, care, maintenance, fit and standards.

Creation Date	04-Oct-2010
Revision Date	11-Sep-2024
Revision Summary	Not applicable.

This safety data sheet complies with the requirements of Regulation (EC) No. 1907/2006. COMMISSION REGULATION (EU) 2020/878 amending Annex II to Regulation (EC) No 1907/2006 .

For Switzerland - Compiled in accordance with the technical provisions referred to in Annex 2, Number 3, ChemO (SR 813.11 - Ordinance on Protection against Dangerous Substances and Preparations).

Disclaimer

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End of Safety Data Sheet